

Homework 4

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1)

1. When c doesn't terminate or errors out.
2. If $\models P \implies Q$.
3. When c terminates.

2)

$$\frac{\{P \wedge b\}c\{Q\} \quad \{P \wedge \neg b\} \text{ skip } \{Q\}}{\{P\}\text{if } b \text{ then } c\{Q\}}$$

We have proven during the exercise session that `if b then c ~ if b then c else skip`. Because the proposed rule is simply a special case of the Hoare rule applied to the equivalent command, it is sound. We could also further simplify the second condition to $\models P \wedge \neg b \implies Q$ using the SKIP and CONSEQUENCE rules.

3)

1. $(z = m \wedge m \leq n) \vee (z = n \wedge n \leq m)$
2. $[(z = m - x \wedge m \leq n \wedge x \leq y) \vee (z = n - y \wedge n \leq m \wedge y \leq x)] \wedge x \geq 0 \wedge y \geq 0$

4)

$$\begin{aligned}
& \{x = i \wedge y = j \wedge 0 \leq i\} \implies \\
& \{x \geq 0 \wedge y = j \wedge 0 = (i - x) \times j\} \\
& z := 0; \\
& \{x \geq 0 \wedge y = j \wedge z = (i - x) \times j\} \\
& \mathbf{while} (x > 0) \mathbf{do} (\\
& \{x \geq 0 \wedge y = j \wedge z = (i - x) \times j \wedge x > 0\} \implies \\
& \{x - 1 \geq 0 \wedge y = j \wedge z + y = (i - x + 1) \times j\} \\
& \quad z := z + y; \\
& \{x - 1 \geq 0 \wedge y = j \wedge z = (i - x + 1) \times j\} \\
& \quad x := x - 1 \\
& \{x \geq 0 \wedge y = j \wedge z = (i - x) \times j\} \\
&); \\
& \{x \geq 0 \wedge y = j \wedge z = (i - x) \times j \wedge \neg(x > 0)\} \implies \\
& \{x = 0 \wedge 0 = 0 \wedge z = i \times j\} \\
& y := 0 \\
& \{x = 0 \wedge y = 0 \wedge z = i \times j\}
\end{aligned}$$

5)

If we define $I \triangleq (b \implies P) \wedge (\neg b \implies Q)$, we can rewrite the rule using logical equivalences as follows

$$\frac{\vdash \{I \wedge b\} c \{I\}}{\vdash \{I\} \mathbf{while} b \mathbf{do} c \{I \wedge \neg b\}}$$

which is the classic WHILE Hoare rule. Because Hoare rules are relatively complete, the equivalent WHILE-ALT rule is too.